Check of your `BFKL.tex`

Notation as in your file: $\mathcal{K}(X) = X/X^2$, $U^{bc} = U^{cb}$. Blue: new or corrected; red: removed. Reference: the log $\frac{\vee}{k^+}$ list of every row in `Checks_JIMWLK_BFKL.pdf` (eq. (B:...)), which comes from the exact split of your current Group I–III rows, and BFKL⊗LO of App. E (Eq. (LOBFKL)).

Method. Every equation is evaluated pointwise: random transverse points, random classical charges, $U = U^T$, $\log \frac{\vee}{k^+} = 1$, and $\epsilon = 0.37$ kept finite in the pole brackets.

- Three-ρ first forms are compared with the reference after symmetrization over the charges (with the c.c. for Groups I, III).
- The "This can be written as" lists are compared with the two-ρ (commutator) terms of the first form, computed directly. With the charges as operators, a product ABC of three charges is its symmetric part plus $\frac{1}{2}([A,B]C+[B,C]A+[A,C]B)$, with $[\rho^a(x), \rho^b(y)] = if^{abc}\rho^c(x)\,\delta^{(2)}(x-y)$; anticommutators are kept ordered, $\{A,B\} = AB+BA$, and the c.c. reverses the order.
- This rule was checked in two ways. It reproduces the reordering engine used before term by term (6 rows, $\leq 2\times10^{-16}$). It also reproduces the checked two-ρ lists of `Evolution_terms.tex` from the log $\frac{k^+}{\Lambda}$ three-ρ terms (Group II Row I, Group III Rows I, II: exact).
- Index sums (δ^{ij}, $(x-w)^i\ldots$) are done term by term. A term without the index is not summed over it.

1 Answer

- **Your result is correct.**
  - The sum of all two-ρ lists of the file is BFKL⊗LO exactly (1.2×10^{-15}, two independent evaluators).
  - Both forms of your "BFKL contribution" are BFKL⊗LO (2.2×10^{-15}), and they equal that sum.
  - The colour-singlet form is unchanged from the version checked before; it equals Eq. (LOBFKL) of App. E.
- **Row by row the file is correct, with three corrections (C1–C3):**
  - every first form equals the log $\frac{\vee}{k^+}$ terms of its row (Table 1), except Group I Row II (C1);
  - every "written as" list is exactly the two-ρ part of its first form (Table 2), Group I Row II with the corrected first form;
  - the two-ρ lists "From 4ρ's" are identical to those of `Evolution_terms.tex` (difference 0).

2 Corrections

C1 Group I, Row II, first form. The equation as written is a copy of Group I Row III, d^3N_1 (identical text). The log $\frac{\vee}{k^+}$ terms of Row II are (`Checks_JIMWLK_BFKL.pdf`, eq. (B:G1II2a)):

$$\begin{aligned}
\frac{d^3N_{2\,a}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\\
&\times\left[\frac{(x-w)^i}{(x-w)^2}\,\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}-\frac{(y-w)^i}{(y-w)^2}\,\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\right]\\
&\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^4}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x'}\frac{(x'-w')^i}{(x'-w')^2}\\
&\times\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2}-\frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]\\
&\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big]. \qquad (1)
\end{aligned}$$

The logarithmic term has no z left, so its measure is $\int_{x,y,x'}$. Your "This can be written as" list for Row II is exactly the two-ρ part of these terms (1.9×10^{-15}), so only the displayed first form changes.

C2 Group II, Row IV, third line (H_4): the sign at the start of the line is missing; it is $+$. Without it LaTeX prints the line as a product. It was also missing in my `Checks_JIMWLK_BFKL.pdf`; corrected there.

C3 Colour-singlet form: the measure is $\int_{x,y,z,w,w'}$ (y is used, there is no x').

3 Row by row

your row (first form)	reference	relative difference
Group I, Row II	(B:G1II2a)	different (C1); corrected: 0
Group I, Row III, d^3N_1, d^3N_2	(B:G1IIIN1), (B:G1IIIN2)	0, 0
Group I, Row IV	(B:G1IV)	0
Group II, Row I, first part (two equations)	(B:G2I1A), (B:G2I1B)	0, 0
Group II, Row I, second part (five equations)	(B:G2I2b)–(B:G2I2f)	0
Group II, Row IV	(B:G2IV)	0 (with C2)
Group III, Rows I, II, III (both parts), V, VI	(B:G3I), ..., (B:G3VI)	0

Table 1: First forms against the log $\frac{\vee}{k^+}$ lists (pointwise; the text is identical up to spacing in every row but Group I Row II).

row	"written as" list against	relative difference
Group I, Row II	two-ρ part of the corrected first form	1.9×10^{-15}
Group I, Row III	two-ρ part of $d^3N_1 + d^3N_2$	1.0×10^{-15}
Group I, Row IV	two-ρ part of the first form	6.0×10^{-15}
Group II, Row I, first part	two-ρ part of the first form	1.8×10^{-14}
Group III, Row I	two-ρ part of the first form	4.4×10^{-15}
Group III, Row II	two-ρ part of the first form	2.9×10^{-15}
Group III, Row III, first part	two-ρ part of the first form	7.5×10^{-16}
Group III, Row V	two-ρ part of the first form	4.4×10^{-16}
Group III, Row VI	two-ρ part of the first form	4.8×10^{-14}
Group II, Row I, second part	the five first-form equations	1.9×10^{-14}
Group II, Row IV	the first form (the first two lines cancel; H_4 stays)	8.8×10^{-14}
Group III, Row III, second part	the first form (with its c.c.)	1.6×10^{-14}
From 4ρ's, d^3N_1, d^3N_2	the same lists in `Evolution_terms.tex`	0, 0
sum of all two-ρ lists	BFKL⊗LO	1.2×10^{-15}
BFKL contribution, first / collected form	BFKL⊗LO	2.2×10^{-15} / 2.2×10^{-15}

Table 2: The "written as" lists. Groups I and III include their c.c. (your lists are written with it).

4 Reproducing

In `nlo_gluon_paper/bfkl_tex_check/`:

```
python3 rerun_first.py      # first forms vs log(vee/k+) lists -> cmp_first.json
python3 cmp_two_rows.py     # two-rho rows: written vs first -> cmp_two_rows.json
python3 run_reorder.py      # three-rho rows: written vs two-rho part -> cmp_reorder.json
python3 rerun_rows.py       # G1 IV, G2 I, G3 I, G3 II  -> cmp_reorder_fixed.json
python3 two_total.py        # all two-rho lists vs BFKL x LO -> two_total.json
python3 rule_vs_engine.py   # ordering rule vs the engine   -> rule_vs_engine.json
python3 test_tool.py; python3 test_tool2.py   # rule on the J terms
python3 make_corrected.py; python3 make_corrected.py marked   # corrected/
```

`reorder.py` computes the two-ρ part; `texfix.py` does the index sums term by term.